User Manual for DIYmalls 3.5" ESP32-S3 IPS Display Capacitive Touchscreen

Model: JC3248W535C

Product Overview

The DIYmalls JC3248W535C is a 3.5-inch IPS capacitive touchscreen module, featuring the ESP32-S3-WROOM-1 as its main control unit. This versatile display supports various functions including displaying photos, playing music and videos, setting timers, and showing real-time clock and weather information. Its capacitive touch interface provides intuitive interaction.



Figure 1: The DIYmalls 3.5-inch ESP32-S3 IPS Display Capacitive Touchscreen module, shown with the various connection cables included in the package.

What's in the Box

- 1x 3.5" ESP32-S3 Display Capacitive Touchscreen Module (Model: JC3248W535C)
- Assorted connection cables (as shown in Figure 1)

Note: A USB-C cable and other accessories are not included and must be supplied separately.

Setup Guide

1. Power Connection

To ensure stable operation, it is recommended to power the display via the **Type-C port**. Use a reliable USB-C cable and a suitable power source (e.g., 5V USB adapter).



Figure 2: Rear view of the module highlighting the Type-C power port, various IO ports, and the battery interface.

2. TF Card Insertion

The module includes a TF card interface for expandable storage. To insert a TF card:

1. Locate the TF card slot on the back of the module.
2. Gently push the TF card into the slot until it clicks into place. Ensure the card is oriented correctly.



Figure 3: Proper method for inserting a TF card into the module. TF card is not included.

3. Cable Connections

Connect the provided cables to their respective ports on the module as needed for your application. These include 1.0mm 4p IO cables, speaker cables, and 1.25mm 4p power cables.

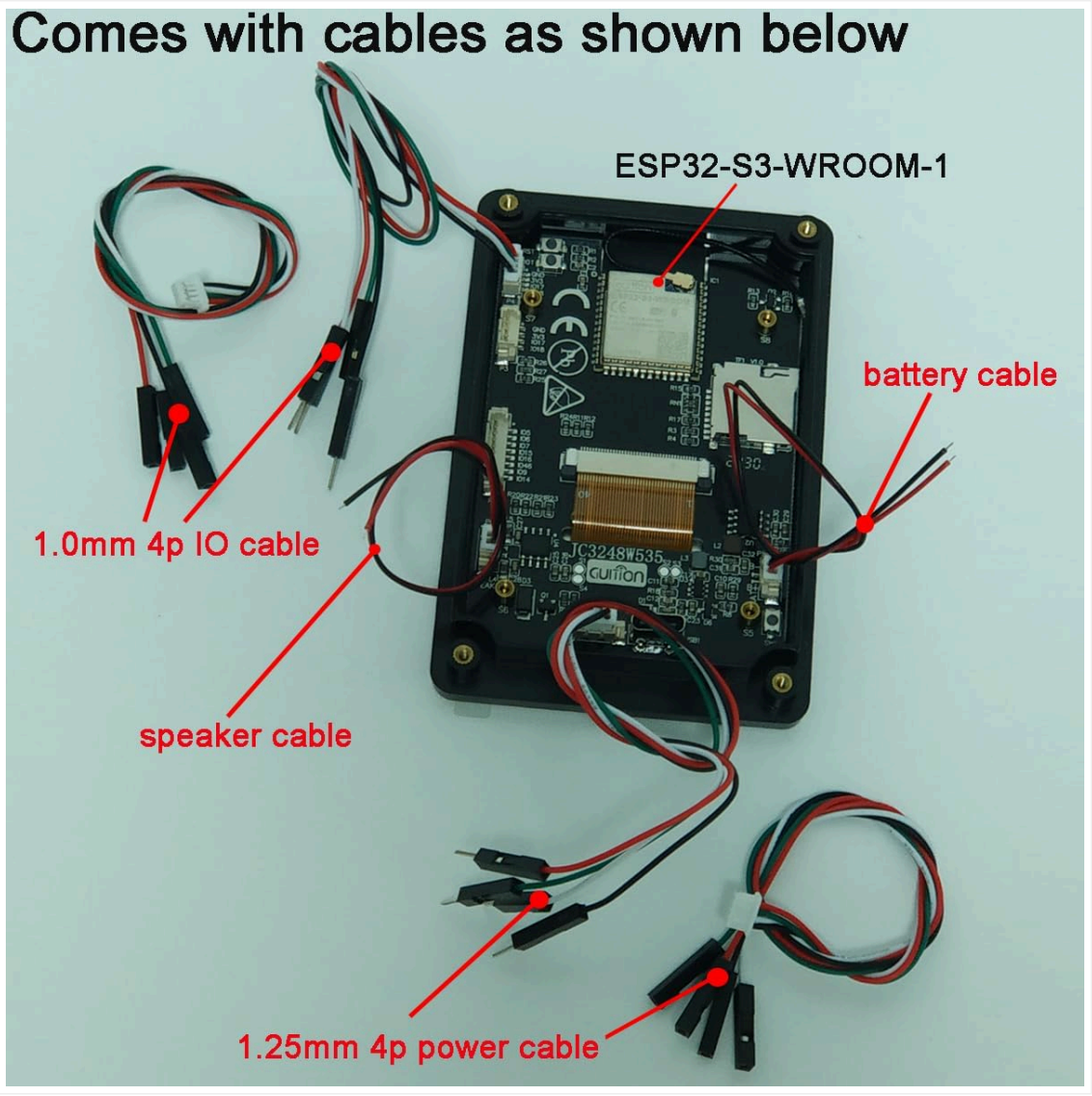


Figure 4: Overview of the included cables and their typical connection points on the ESP32-S3 module, including battery and speaker connections.

4. Case Installation

The module can be secured within a case using screws. Ensure proper alignment before fastening.



Figure 5: Illustration of how the module fits into its case, secured by screws.

5. Uploading Code

To program the ESP32-S3 module, you can upload code using an Integrated Development Environment (IDE) like Arduino IDE or a dedicated ESP32 tool. Connect the module to your computer via the Type-C USB port.

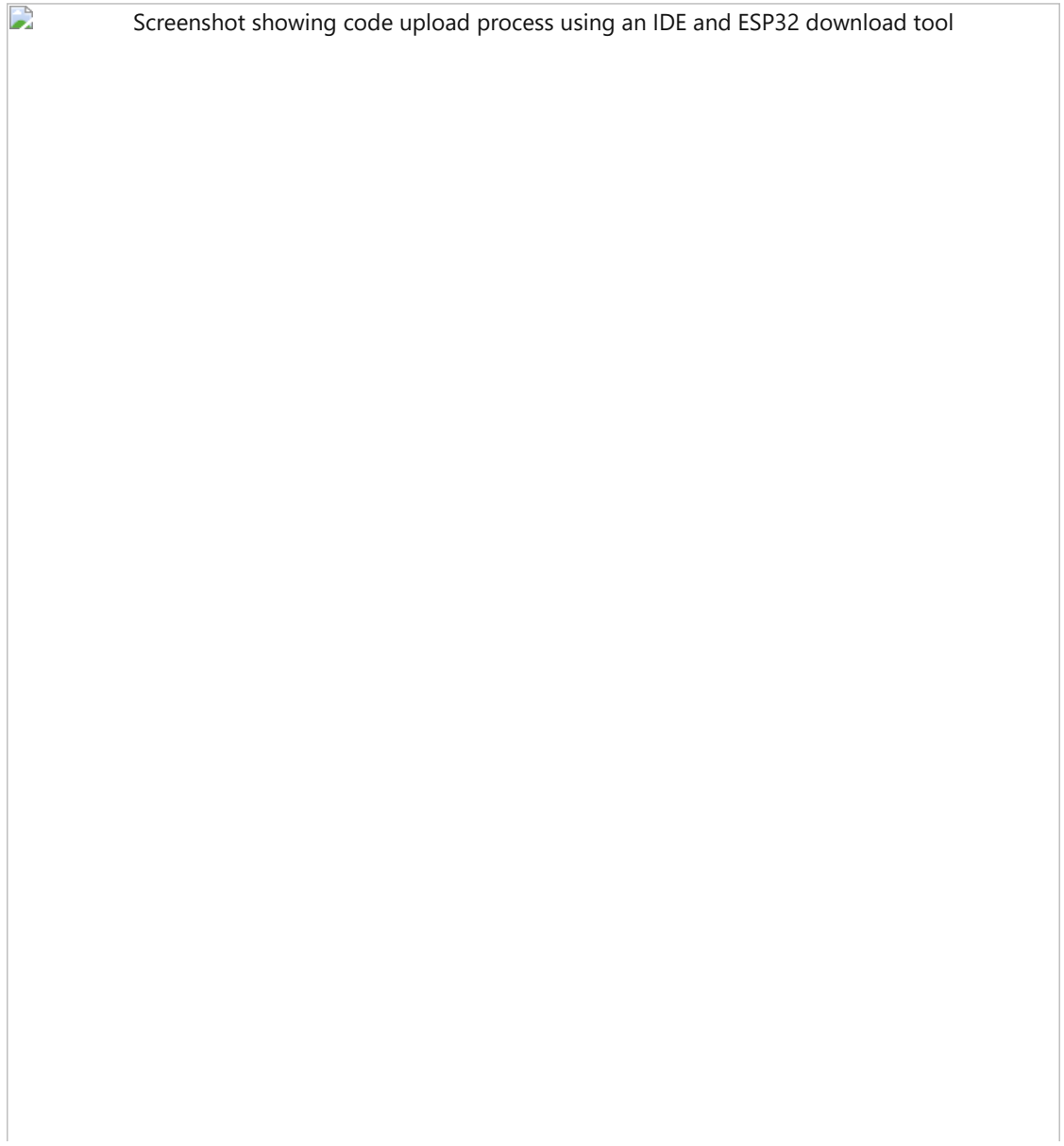


Figure 6: Example of uploading code to the ESP32-S3 module using a development environment.

Operating Instructions

Once powered and programmed, the 3.5-inch capacitive touchscreen can perform various functions:

- **Display Photos:** View images stored on the device or connected storage.
- **Play Music:** If a speaker is connected, the module can play audio files.
- **Play Video:** Display video content.
- **Set Timer:** Utilize the built-in timer function.
- **Real-time Clock:** Show the current time.
- **Weather Information:** Display weather updates (requires internet connectivity and appropriate programming).

Multimedia Playback

The display supports playback of music and video files. Ensure compatible file formats are used.

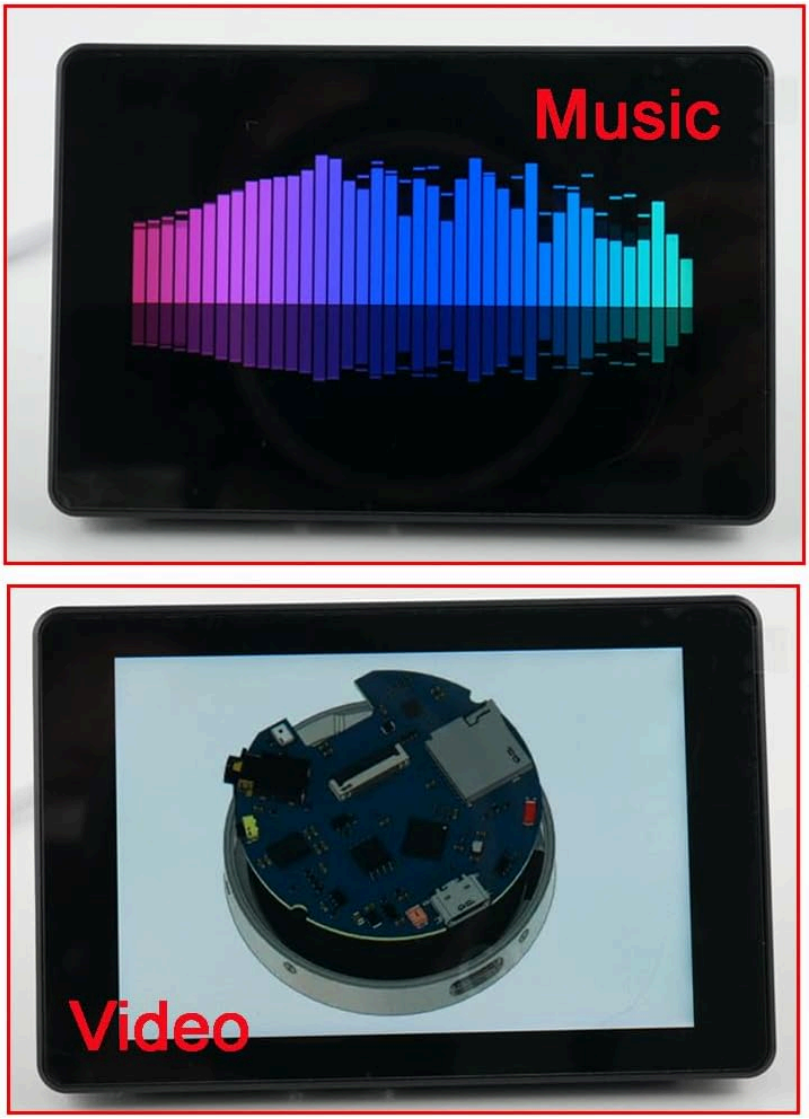


Figure 7: Examples of music playback (top) and video playback (bottom) on the display.



Play video

Video 1: A demonstration of the DIYmalls 3.5" ESP32-S3 Touch Screen Interface in operation, showcasing its capabilities.

Timer and Clock Functions

The module can be programmed to function as a timer or a real-time clock.

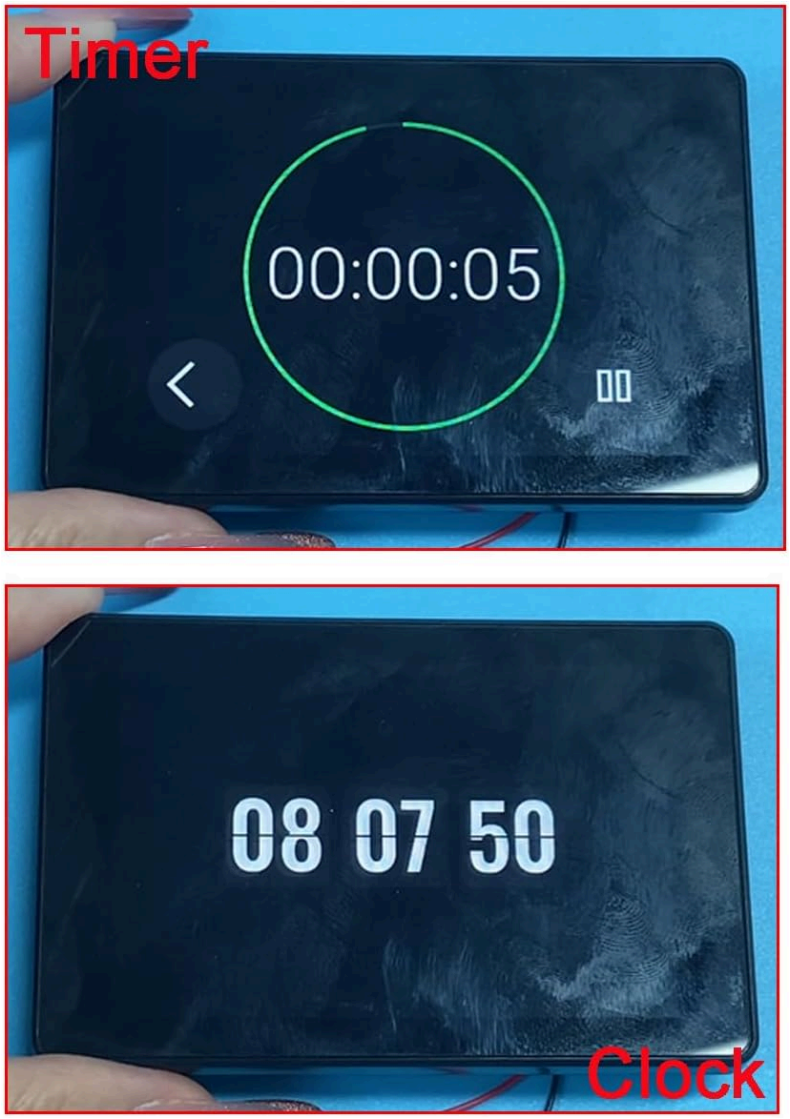


Figure 8: Examples of the timer function (top) and digital clock display (bottom).

Maintenance

- Keep the module clean and free from dust and moisture.
- Avoid exposing the device to extreme temperatures or direct sunlight.
- Always use the recommended 5V power supply via the Type-C port to prevent damage.
- Handle the touchscreen with care to avoid scratches or impact damage.

Troubleshooting

Problem	Possible Cause	Solution
Device does not power on.	Insufficient or no power supply.	Ensure the Type-C cable is securely connected and the power source provic a different cable or power adapter.
Demo application does not work as expected.	Pre-loaded firmware issues or incorrect setup.	It may be necessary to set up a programmer, wipe the device, and re-uploa application or your custom code. Refer to programming guides for ESP32-S
Cannot connect to Wi-Fi.	Incorrect Wi-Fi credentials or network configuration.	Ensure correct Wi-Fi SSID and password are provided in your code. If a defa required for initial setup, contact DIYmalls support for specific details.
Display is blank or shows artifacts.	Loose display ribbon cable or software issue.	Gently check the ribbon cable connection to the display. Re-upload your co software integrity.

If you encounter issues not listed here or require further assistance, please refer to the Support section.

Specifications

Feature	Detail
Display Color	16 BIT RGB 65K color
Size	3.5 inch
View	IPS
Touch Type	Capacitive Touch
Display Type	TFT
Driver Chip	AXS15231B
Resolution	320x480 (Pixel)
Effective Display Area	73.4 x 49.0 mm
Module Size	94.5 x 62.0 mm
Operating Voltage	5V
Main Control Module	ESP32-S3-WROOM-1
Item Weight	3.87 ounces
Item Model Number	JC3248W535C